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## Genetic identification of an expansive mechanoreceptor sensitive to skin stroking

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### Summary

Touch perception begins with activation of low-threshold mechanoreceptors (LTMRs) in the periphery. LTMR terminals exhibit tremendous morphological heterogeneity that specifies their mechanical receptivity. In a survey of mammalian skin, we found a preponderance of neurofilament-heavy chain<sup>+</sup> circumferential endings associated with hair follicles, prompting us to develop a genetic strategy to interrogate these neurons. Targeted *in vivo* recordings revealed them to be A $\beta$  Field-LTMRs, identified 50 years ago but largely elusive thereafter. Remarkably, while A $\beta$  Field-LTMRs are highly sensitive to gentle stroking of the skin, they are unresponsive to hair deflection, and they encode skin indentation in the noxious range across large, spotty receptive fields. Individual A $\beta$  Field-LTMRs form up to 180 circumferential endings, making them the most anatomically expansive LTMR identified to date. Thus, A $\beta$  Field-LTMRs are a major mammalian LTMR subtype that forms circumferential endings in hairy skin, and their sensitivity to gentle skin stroking arises through integration across many low-sensitivity circumferential endings.

### Introduction

Sensitivity to mechanical stimuli is crucial for communication and survival of all organisms. In the case of mammalian skin, innocuous mechanical stimuli are detected by a diverse group of cutaneous low-threshold mechanoreceptor (LTMR) subtypes with distinct morphologies, physiological properties and functions. LTMRs are pseudo-unipolar sensory

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#### Author Contributions

L.B. developed genetic labeling strategies and performed the anatomical characterization with assistance from T.L.D. and P.N.; B.P.L. performed electrophysiological and mechanical measurements with assistance from N.L.N.; J.L. performed FEM simulations; C.J.W. and C.C. performed the *ex vivo* recording experiments. L.B., B.P.L., and D.D.G. wrote the paper with input from all authors.

neurons whose cell bodies reside within dorsal root ganglia (DRG) or trigeminal ganglia. Their touch sensitivity derives from mechanosensitive peripheral axonal terminals that reside in the skin, where they associate with end organs that determine the geometry of their axonal terminals and force sensitivity (Loewenstein, 1969; Zimmerman et al., 2014). Touch stimuli are encoded and further transferred via axons to the central nervous system (CNS), where LTMR inputs carrying distinct aspects of touch information converge and integrate (Abraira and Ginty, 2013; Lechner and Lewin, 2013).

In mammalian hairy skin, at least seven LTMR subtypes tuned to distinct but overlapping features of a tactile stimulus have been described (Abraira and Ginty, 2013; Horsch et al., 1977). Each subtype forms specialized axonal terminals associated with end organs. A $\beta$  RA-LTMRs, A $\delta$ -LTMRs and C-LTMRs form lanceolate endings, which extend along the long axis of hair follicles and are rapidly or intermediately adapting to skin indentation (Li et al., 2011). A $\beta$  SA1-LTMRs form disc-like axonal terminals that associate with Merkel cells, which are located in the basal epidermis around guard hair follicles and are themselves mechanically sensitive, endowing A $\beta$  SA1-LTMRs with their characteristic slowly adapting response property (Ikeda et al., 2014; Maksimovic et al., 2014; Woo et al., 2014). In contrast, A $\beta$  RA2-LTMRs terminate deep in the dermis where they are wrapped by layers of cushion-like lamellar cells within Pacinian corpuscles; these serve as high pass mechanical filters that underlie the unique A $\beta$  RA2-LTMR high frequency vibration tuning property. A $\beta$  SA2-LTMRs also terminate in the dermis and may form Ruffini endings (Chambers et al., 1972). The least understood LTMR subtype, the A $\beta$  Field-LTMR (formerly called field receptor), first described nearly 50 years ago in the cat by Burgess and colleagues, is rapidly or intermediately adapting to skin indentation (Burgess et al., 1968). These neurons are sensitive to gentle stroking of the skin but rarely respond to deflection of individual hairs (Horsch et al., 1977). The terminal structure and skin arborization patterns of A $\beta$  Field-LTMRs have not been determined, leaving open the question of how their distinct physiological properties arise.

All LTMRs have a central axonal projection that innervates the spinal cord dorsal horn. Undirected recordings indicate that five A $\beta$ -LTMR subtypes, including the A $\beta$  Field-LTMR, also have an axonal branch that ascends via the dorsal column (DC) to the dorsal column nuclei (DCN) in the brainstem (Horsch et al., 1976), and these projections are considered important for the perception of discriminative touch (Ballermann et al., 2001; Dobry and Casey, 1972). Here we have taken an unbiased retrograde labeling approach to visualize the cutaneous endings of A $\beta$ -LTMRs with axons that ascend the DC to the DCN of mice. These experiments guided molecular-genetic strategies to label the A $\beta$ -LTMRs for which genetic tools do not currently exist, specifically A $\beta$  SA1-LTMRs as well as large myelinated cutaneous sensory neurons that give rise to circumferential endings associated with hair follicles. Importantly, physiological recordings from genetically labeled circumferential ending neurons reveal them to display key hallmarks of the elusive A $\beta$  Field-LTMRs. We found that in comparison to other hairy skin LTMRs, this LTMR subtype exhibits striking peripheral receptive field properties, unique ultrastructural features, characteristic LTMR central innervation patterns, and highly distinct tuning properties. Our findings establish the morphological and physiological properties of the A $\beta$  Field-LTMR and support a model in

which the weak mechanosensitivity and expansive organization of its circumferential endings explain this neuron's sensitivity to gentle stroking of the skin.

## Results

### Identification of cutaneous sensory neurons that innervate the DCN

To visualize the cutaneous terminal morphologies of uncharacterized DCN-projecting A $\beta$ -LTMR subtypes, we designed an unbiased strategy to selectively label neurons exhibiting direct DCN projections. AAV2/1-Cre virus was injected bilaterally into the rostral DC at the first cervical level of *R26<sup>LSL-tdTomato</sup>* mice to retrogradely infect DRG neurons and visualize their cutaneous axonal projections (Figure 1A). TdTomato<sup>+</sup> DRG neurons have large cell bodies and are neurofilament-heavy chain<sup>+</sup> (NFH<sup>+</sup>) (Figure S1A), but they do not bind to the lectin IB4 and very few express CGRP (Figures S1C and S1D). These findings indicate that our retrograde labeling strategy is specific because small or medium sized DRG sensory neurons, including C-LTMRs, A $\delta$ -LTMRs and most nociceptors, do not have an axonal branch that projects directly to the brainstem.

We next visualized the cutaneous terminal morphologies of tdTomato<sup>+</sup> neurons. In hairy skin, tdTomato<sup>+</sup> neurons form four types of endings. Close to the epidermis and between hair follicles, tdTomato<sup>+</sup> axonal terminals were observed wrapped by S100<sup>+</sup> Schwann cells (Figure 1B) in structures that resemble Pacinian corpuscles (Luo et al., 2009); however, unlike Pacinian corpuscles, these tdTomato<sup>+</sup> endings do not reside within the deep dermis. This is a curious neuronal population with unknown function. The other three types of tdTomato<sup>+</sup> axons observed form terminals associated with hair follicles. At the neck of guard hair follicles, tdTomato<sup>+</sup> axonal endings associate with Troma1<sup>+</sup> Merkel cells (Figure 1C); these tdTomato<sup>+</sup> neurons are A $\beta$  SA1-LTMRs. A third population forms NFH<sup>+</sup> lanceolate endings (Figure 1D); these neurons are A $\beta$  RA-LTMRs. A fourth class of tdTomato<sup>+</sup> DRG neurons displays circumferential endings that also encircle hair follicles below the sebaceous gland (Figure 1D). Similar to A $\beta$  SA1-LTMR and A $\beta$  RA-LTMR endings, the majority of tdTomato<sup>+</sup> circumferential endings are NFH<sup>+</sup> (Figure 1C-1D), suggesting these neurons have myelinated axons with conduction velocities (CVs) in the A $\beta$  range. Whole-mount staining of back hairy skin revealed that NFH<sup>+</sup> circumferential endings innervate 94% of hair follicles, whereas A $\beta$  RA-LTMRs and A $\beta$  SA1-LTMRs innervate only 8% and 0.8% of hair follicles, respectively (Figure 1E). Thus, we have identified a DCN-projecting DRG neuronal population with a striking preponderance of NFH<sup>+</sup> circumferential endings in hairy skin of mice.

Circumferential endings were first described in rodents and cats in the 1980s (Millard and Woolf, 1988; Rice and Munger, 1986). Just as there are multiple sensory neuron subtypes with lanceolate endings, experiments examining expression patterns of molecular markers showed there are multiple neuronal types with circumferential endings (Fünfschilling et al., 2004). Indeed, we found that in mouse hairy skin, there are only two molecularly distinct types of circumferential endings surrounding hair follicles, one expressing CGRP and the other expressing NFH (Figure 1E, S1E-S1H). Furthermore, all circumferential endings labeled by DCN retrograde labeling are NFH<sup>+</sup> and CGRP<sup>-</sup> (Figure 1D and S1I); thus, only neurons with NFH<sup>+</sup> circumferential endings project directly to the DCN. In a multi-species

survey of cutaneous sensory neurons, we found that NFH<sup>+</sup> circumferential endings are also present in cats, dogs, and macaques (Figure 1F-I), indicating that these neurons are prevalent amongst mammals.

### Intersectional genetic strategies to selectively label neurons with NFH<sup>+</sup> circumferential endings and A $\beta$ SA1-LTMRs

The anatomical, electrophysiological, and functional properties of DRG neurons with circumferential endings surrounding hair follicles are unknown. We therefore sought to generate molecular genetic tools that enable investigation of these neurons. NT3-TrkC signaling mediates development of NFH<sup>+</sup> circumferential endings (Albers et al., 1996) and half of DCN-projecting DRG neurons are TrkC<sup>+</sup> (Figure S1B). Moreover, TrkC<sup>+</sup> DRG neurons of adult mice can be subdivided into three mutually exclusive populations: Ret<sup>+</sup> neurons; Parvalbumin<sup>+</sup> (PV<sup>+</sup>) proprioceptors; and CGRP<sup>+</sup> peptidergic nociceptors (Figure S3A and S3B). Thus, we tested the hypothesis that the TrkC<sup>+</sup>/Ret<sup>+</sup> population includes neurons with NFH<sup>+</sup> circumferential endings.

We first generated a *TrkC<sup>tdTomato</sup>* knock-in mouse line (Figure S2A and S2B), in which the pattern of tdTomato expression recapitulates that of TrkC (Figure S2C). In hairy skin, tdTomato is found in all NFH<sup>+</sup> circumferential endings (Figure S2D), all A $\beta$  SA1-LTMR endings associated with Merkel cells, and a subset of free nerve endings (Figure S2E). Importantly, the NFH<sup>+</sup> circumferential endings and Merkel endings labeled by *TrkC<sup>tdTomato</sup>* are also Ret<sup>CFP+</sup> (Figure S2F, S3C, and S3D).

To selectively label the TrkC<sup>+</sup>/Ret<sup>+</sup> DRG neuronal populations, we designed an intersectional genetic labeling strategy. For this, we generated a *TrkC<sup>CreER</sup>* knock-in mouse line (Figure S2A and S2B). When *TrkC<sup>CreER</sup>* mice were crossed with *Ret<sup>f(CFP)</sup>* mice (Uesaka et al., 2008), which express CFP under the control of the endogenous *Ret* locus following Cre-mediated recombination (Figure 1J), tamoxifen treatment promoted CFP expression exclusively in TrkC<sup>+</sup> cells (Figure S3F and S3I). Interestingly, we found that treatment of *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* mice with 0.1 mg of tamoxifen at P5 led to expression of CFP in the majority of NFH<sup>+</sup> circumferential endings (Figure 1K and 1M), but not in other populations, including neurons with Merkel endings, lanceolate endings, free nerve endings, or expressing CGRP<sup>+</sup> or PV<sup>+</sup> (Figure 1K, 1M, S3G, and S3H). Based on the efficiency of terminal labeling and the percentage of labeled NeuN<sup>+</sup> DRG neurons, we estimate that neurons with NFH<sup>+</sup> circumferential endings represent 4.3  $\pm$  0.3% of all thoracic DRG neurons (3853 neurons counted in 4 mice), which is comparable to the density of A $\beta$  RA-LTMRs and A $\delta$ -LTMRs (Li et al., 2011).

In contrast, treatment of *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* mice with 3 mg of tamoxifen at E12.5 led to expression of CFP exclusively in Merkel endings that are derived from A $\beta$  SA1-LTMRs (Figure 1L and 1M), but not in other populations (Figure 1L, 1M, S3J, and S3K). Thus, the tamoxifen-inducible *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* intersectional genetic strategy enables selective labeling of either neurons with NFH<sup>+</sup> circumferential endings or A $\beta$  SA1-LTMRs, depending on the time and dosage of tamoxifen delivery.

## Neurons with NFH<sup>+</sup> circumferential endings project to the deep spinal cord dorsal horn and the DCN

We next used the *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* intersectional genetic labeling strategy to visualize the central projections of neurons with NFH<sup>+</sup> circumferential endings. In the spinal cord, these CFP<sup>+</sup> neurons project within lamina II<sub>IV</sub> to lamina IV, partially overlapping with the PKCγ<sup>+</sup> interneurons of lamina II<sub>IV</sub> and III but not with IB4<sup>+</sup> terminals in lamina II<sub>id</sub> (Figure 2A-2B). These spinal cord terminations overlap with, but are slightly more dorsal than those of other Aβ-LTMR subtypes, including Aβ RA-LTMRs (Luo et al., 2009) and Aβ SA1-LTMRs that innervate laminae III to V (Figure 2C-2D), and they resemble the spinal cord innervation patterns of a subset of myelinated nociceptors (Boada and Woodbury, 2008). Furthermore, CFP<sup>+</sup> circumferential ending neurons have central projections that terminate in the gracile and the dorsal cuneate nuclei of the DCN (Figure 2F-2G), but are excluded from the ventral cuneate and external cuneate nuclei (Figure 2E-2G), similar to Aβ RA-LTMRs (Luo et al., 2009).

In order to visualize the central projections of individual NFH<sup>+</sup> circumferential ending neurons, we next used a sparse genetic labeling strategy, employing *TrkC<sup>CreER</sup>* in combination with *Brn3a<sup>f(AP)</sup>*, a Cre-dependent alkaline phosphatase (AP) reporter that is expressed in most NFH<sup>+</sup> DRG neurons but not in the spinal cord (Badea et al., 2012). The central projections of individual NFH<sup>+</sup> circumferential ending neurons bifurcate upon innervating the spinal cord, then project in both the rostral and caudal directions and sprout multiple collaterals that span  $3.7 \pm 0.3$  spinal segments (Figure 2L and S4A). Moreover, the rostral axonal branch of each NFH<sup>+</sup> circumferential ending neuron projects to the DCN where it forms collaterals (Figure 2I and S4A). These central projection patterns resemble hairy skin Aβ-LTMR subtypes including Aβ RA-LTMRs (Niu et al., 2013) and Aβ SA1-LTMRs, which form collaterals spanning  $3.9 \pm 0.2$  spinal segments (Figure 2M and S4B) and innervate the DCN (Figure 2K) although, interestingly, the majority of Aβ SA1-LTMRs in DRGs below T10 do not project to the DCN (Figure S4B). These anatomical findings implicate DRG neurons with NFH<sup>+</sup> circumferential endings in tactile perception.

## Neurons with NFH<sup>+</sup> circumferential endings are Aβ Field-LTMRs sensitive to gentle stroking of the skin but are relatively insensitive to skin indentation

In order to evaluate the physiological properties of NFH<sup>+</sup> circumferential ending neurons, we developed a targeted *in vivo* electrophysiological recording paradigm (Figure 3A). We crossed *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* with a dual recombinase dependent reporter line *R26<sup>LSL-FSF-tdTomato</sup> (Ai65)* and subcutaneously injected an AAV2/1-Flpo virus to label TrkC<sup>+</sup> neurons innervating the dorsal surface of the thigh. This allowed us to confine our recordings to NFH<sup>+</sup> circumferential neurons innervating a small region of hairy skin amenable to mechanical stimulation. Recently, NFH<sup>+</sup> circumferential ending neurons were suggested to function as Aδ-mechanonociceptors on the basis of *ex vivo* recordings from a mixed population of DRG neurons labeled by *DOR<sup>GFP+</sup>* (Bardoni et al., 2014). Our intersectional genetic strategy allowed us to unambiguously and quantitatively explore the link between the structure and function of NFH<sup>+</sup> circumferential ending neurons.

We first assessed responsiveness of NFH<sup>+</sup> circumferential ending neurons to skin stroke. Upon gentle stroking of the skin in the rostral to caudal direction with a fine brush (average holding force of 5 mN, 1-10 mm/sec sweep velocity), NFH<sup>+</sup> circumferential ending neurons showed robust responses with an action potential firing frequency that equaled or exceeded those of A $\beta$  RA-LTMRs and A $\beta$  SA1-LTMRs labeled using *Npy2r-GFP* mice and *TrkC<sup>CreER</sup>; Ret<sup>fl(CFP)</sup>* mice treated with tamoxifen at E12.5, respectively (Figure 3B-3C). Moreover, electrical stimulation of the skin demonstrated that NFH<sup>+</sup> circumferential ending neurons have a CV of  $17.0 \pm 2.0$  m/sec, which is indistinguishable from the CVs of A $\beta$  RA-LTMRs and A $\beta$  SA1-LTMRs, respectively (Figure 3D). Thus, NFH<sup>+</sup> circumferential ending neurons are A $\beta$ -LTMRs highly sensitive to gentle skin stroke.

Interestingly, NFH<sup>+</sup> circumferential ending neurons exhibit a much higher force threshold than A $\beta$  RA-LTMRs and SA1-LTMRs when the skin is indented with von Frey filaments (Figure 3E). These responses exhibited considerable trial-to-trial variability; therefore, we next used a force-controlled stimulator which could be precisely and consistently positioned over the skin. We found that each NFH<sup>+</sup> circumferential ending neuron's receptive field displayed "hotspots", which we subsequently targeted to assess indentation responses. NFH<sup>+</sup> circumferential ending neurons exhibited distinct adaptation properties compared to A $\beta$  SA1-LTMRs and A $\beta$  RA-LTMRs. At forces near threshold their responses adapted rapidly over tens of milliseconds, while at higher forces adaptation responses to indentation steps became intermediate between A $\beta$  SA1-LTMRs and A $\beta$  RA-LTMRs (Figure 3F). NFH<sup>+</sup> circumferential ending neurons also show few or no off responses, which are a hallmark of A $\beta$  RA-LTMRs (Figure 3F), further demonstrating this is a distinct LTMR population.

We also made intracellular recordings from NFH<sup>+</sup> circumferential neurons using an *ex vivo* preparation. These recordings revealed inflected somal action potentials that were broader than the uninflected somal action potentials of A $\beta$  RA-LTMRs (Figure S5A). Consistent with the *in vivo* recordings, individual receptive fields were composed of multiple hotspots. Application of controlled forces using a blunt probe (1 mm diameter) centered over a hotspot revealed that NFH<sup>+</sup> circumferential neurons have an indentation threshold about 5 mN (Figure S5C), which is considerably higher than that of both A $\beta$  RA-LTMRs and A $\beta$  SA1-LTMRs (McIlwrath et al., 2007). Moreover, indentation responses became more slowly adapting and firing rates increased as forces increased into the noxious range (Figure S5C). These neurons responded far more vigorously when the same hotspots were indented with sharp probes (Figure S5D), and even more robustly when coaxial forces were applied to individual hairs (hair pull) in the hotspot (Figure S5B and S5E). Thus, while NFH<sup>+</sup> circumferential neurons are highly responsive to gentle skin stroking, and are therefore an LTMR, these neurons also exhibit hallmarks of myelinated nociceptors (Burgess and Perl, 1967; Djouhri and Lawson, 2004; Koerber et al., 1988).

Both A $\beta$  RA-LTMRs and A $\beta$  Field-LTMRs are hairy skin innervating, predominantly rapidly adapting LTMR subtypes that are sensitive to skin stroking. Unlike A $\beta$  RA-LTMRs, A $\beta$  Field-LTMRs do not respond to deflection of individual guard hairs (Horch et al., 1977). Therefore, we next performed *in vivo* recordings to ask whether NFH<sup>+</sup> circumferential ending neurons are sensitive to guard hair deflection by moving a brush across the tips of guard hairs, which extend beyond awl/auchene and zigzag hairs. In all cases tested, this

stimulus elicited bursts of spikes in A $\beta$  RA-LTMRs but failed to excite NFH<sup>+</sup> circumferential ending neurons (Figure 3G and 3H). Taken together, the NFH<sup>+</sup> circumferential ending neuron is an A $\beta$ -LTMR that is exquisitely sensitive to gentle stroking of the skin, relatively insensitive to skin indentation, with a rapidly adapting discharge at low to moderate indentation forces, and insensitive to deflection of guard hairs. Thus, the properties of NFH<sup>+</sup> circumferential ending neurons match those of both classically defined A $\beta$  Field-LTMRs (Horch et al., 1977) as well as a subset of myelinated nociceptors (Burgess and Perl, 1967), and are hereafter referred to simply as A $\beta$  Field-LTMRs.

### **A $\beta$ Field-LTMRs associate with hair follicles yet are insensitive to hair deflection**

A $\beta$  Field-LTMRs reportedly respond to deflection of multiple hairs spanning a large stimulus “field”, which may be explained by simultaneous deflection of many hairs or the inevitable stimulation of skin that occurs when stroking a large group of hairs (Horch et al., 1977). To distinguish between these possibilities, we measured the extent of skin and hair displacement orthogonal to the skin surface and found that air puff stimuli could be titrated to elicit varying amounts of hair deflection without significant skin displacement, while indentation with a servo-actuated von Frey filament produced displacement of both hairs and skin that were similar in magnitude (Figure 4A and 4B).

Next, we measured the response properties of A $\beta$  Field-LTMRs to air puff-induced hair deflection. As expected, both air puff and skin stroking elicited robust spiking in A $\beta$  RA-LTMRs (Figure 4C). Moreover, A $\beta$  RA-LTMR spike frequency increased with the speed of the air puff, and reached a plateau at an air puff speed of 30 m/sec (Figure 4D). Remarkably, the response of A $\beta$  RA-LTMRs to air puff was entirely dependent on the presence of hairs; experiments in which hairs were removed during the recording period showed that the response to air puff, but not to skin stroking, was abolished following hair removal (Figure 4C and 4D), further confirming that air puff elicits hair deflection with minimal skin indentation. In contrast, neither A $\beta$  Field-LTMRs nor A $\beta$  SA1-LTMRs responded to air puff at any stimulus intensity, including those that saturate the A $\beta$  RA-LTMRs (Figure 4C and 4D). Moreover, A $\beta$  Field-LTMR responses to gentle stroking of the skin were similar before and after removing hairs, demonstrating that the stroke sensitivity of these neurons does not rely on hair shafts. Thus, A $\beta$  Field-LTMRs are categorically insensitive to hair deflection despite their association with hair follicles. Moreover, A $\beta$  Field-LTMRs, A $\beta$  RA-LTMRs and A $\beta$  SA1-LTMRs have distinct response properties to mechanical stimulation; while all three A $\beta$ -LTMR subtypes respond to gentle stroking of the skin, A $\beta$  RA-LTMRs and A $\beta$  SA1-LTMRs but not A $\beta$  Field-LTMRs respond to gentle skin indentation, and only A $\beta$  RA-LTMRs respond to hair deflection.

### **Micromechanical determinants of A $\beta$ -LTMR sensitivity**

A $\beta$  Field-LTMRs and A $\beta$  RA-LTMRs both form endings around hair follicles and yet they exhibit different sensitivities to hair deflection and skin indentation. We hypothesized that the ultrastructural architecture of their cutaneous endings and the mechanics of the surrounding tissues may explain this differential sensitivity. We first used transmission electron microscopy (TEM) to define the ultrastructural features of circumferential and lanceolate terminals. Interestingly, a cross sectional analysis of individual hair follicles

clearly shows that while both lanceolate (A $\beta$  RA-LTMR) and circumferential (A $\beta$  Field-LTMR) endings are associated with hair follicles, they reside within distinct layers surrounding follicles. Lanceolate endings are located within an inner region where they closely abut the basement membrane of hair follicle epithelial cells with an average distance  $\sim 100$  nm (Figure 5A and 5C), as previously described (Li and Ginty, 2014). In contrast, circumferential endings reside within an outer region of the follicle surrounding the lanceolate endings, and are thus considerably more distant from the follicle (Figure 5A-5C; average distance 4  $\mu$ m). Strikingly, both types of sensory endings are aligned with collagen fibers that are organized in parallel to the orientation of the axon terminals (Figure 5A-5C).

To identify an ultrastructural basis for the unique tuning properties of A $\beta$  RA-LTMRs and A $\beta$  Field-LTMRs, we estimated the extent to which hairy skin mechanical stimulation is transformed into strain, a normalized local deformation, acting upon lanceolate and circumferential endings using a finite element model (FEM) encompassing the mechanical properties of the skin, hair, hair follicle and the two layers of collagen networks. We simulated activation of axon terminals by delivering three stimuli that mimic those used for electrophysiological recordings (see Figures 3 and 4): 100  $\mu$ m hair deflection, and 0.8 mN or 10 mN indentation applied to a skin region adjacent to the hair follicle (Figure 5D-5F, bottom). FEM simulations suggest that considerably more strain acts upon lanceolate endings than on circumferential endings in response to both hair deflection and skin indentation (Figure 5D-5H). The maximum strain on lanceolate endings in response to 100  $\mu$ m hair deflection or 0.8 mN skin indentation is comparable ( $\sim 2e^{-3}$ ), and is much higher than that acting on circumferential endings ( $\sim 5e^{-4}$ ) (Figure 5D-5E and 5G-5H). Simulation using 10 mN skin indentation, which is close to the von Frey threshold of A $\beta$  Field-LTMRs, predicted the maximum strain acting on circumferential endings to be  $\sim 6e^{-3}$  (Figure 5F). This is greater than the strain following 100  $\mu$ m hair deflection (18 fold) and 0.8 mN skin indentation (12 fold) but is only slightly higher than the strain on lanceolate endings in response to 100  $\mu$ m hair deflection and 0.8 mN skin indentation (3 fold). These simulated strain measurements are consistent with our physiological measurements of the relative sensitivities of the two neuronal types.

We next performed similar FEM simulations in the absence of the two layers of collagen observed by TEM. The strain on lanceolate endings was similar in the presence and absence of the two layers of collagen. In contrast, for circumferential endings, the strain in response to indentation in the absence of two collagen layers was approximately 50 times greater than that when both collagen layers were present (Figure 5I). Taken together, these FEM simulations suggest that the strain acting on A $\beta$  RA-LTMR lanceolate endings is greater than the strain acting on the A $\beta$  Field-LTMR circumferential endings following both hair deflection and skin indentation, and that the outer collagen layer may serve to dampen responses of A $\beta$  Field-LTMR circumferential endings to skin indentation.

### **A $\beta$ Field-LTMRs have unusually large receptive fields with many weakly mechanosensitive foci, and long distances between their axon terminals and spike initiation segments**

Although A $\beta$  Field-LTMRs are relatively insensitive to focal indentation, they are highly sensitive to gentle stroking of skin. How do A $\beta$  Field-LTMRs acquire their sensitivity to

gentle stroking across large areas of skin? One potential mechanism is that many weakly mechanosensitive endings of individual A $\beta$  Field-LTMRs are elaborated over a large area of skin, and thus stimuli that sweep across the entirety of an A $\beta$  Field-LTMR's endings generate multiple receptor potentials that sum throughout the arbor and induce greater responses. To test this hypothesis, we first generated fine scale receptive field maps of A $\beta$  Field-LTMRs using *in vivo* electrophysiological recordings. Stimuli were delivered using a sharp-tipped (20  $\mu$ m in diameter) force controlled indenter, which was systematically translated over the skin of the dorsal thigh at 50  $\mu$ m intervals. This distance was chosen to match the spatial scale of hair follicles and the spacing between them (Figure S6A and S6B). The resulting spatial receptive field maps are composed of the spiking responses to indentations delivered to 3200 separate positions.

Strikingly, A $\beta$  Field-LTMR responses to 27 mN step indentations showed that the punctuate "hotspots", described above in both *in vivo* and *ex vivo* recordings, are distributed over 3-4 mm<sup>2</sup> of the skin. Responses to stimulation of hotspots (half-width  $\sim$  60  $\mu$ m, n = 2 with full receptive fields) exceeded 100 Hz, and hotspots were separated by insensitive stretches where skin stimulation evoked 0-2 spikes (Figure 6A and 6B). Thus, individual A $\beta$  Field-LTMRs exhibit unusually large receptive fields composed of many weakly mechanosensitive hotspots distributed over a 3-4 mm<sup>2</sup> area of skin.

To compare the physiological receptive fields to the peripheral anatomy of individual A $\beta$  Field-LTMRs, we next performed sparse labeling using *TrkC<sup>CreER</sup>; Brn3a<sup>fl(AP)</sup>* mice. Individual A $\beta$  Field-LTMR axons branch in the skin to give rise to circumferential endings associated with a remarkably large number of hair follicles (Figure 6C, S6C and S6E). Despite a large range in the number of associated hair follicle endings (20-180 hair follicles per neuron), most A $\beta$  Field-LTMRs innervate a large area of skin ( $3.1 \pm 0.1$  mm<sup>2</sup>, Figure 6C, S6D and S6F). Importantly, these A $\beta$  Field-LTMR morphologies bear striking resemblance to their receptive fields (Figure 6B), suggesting that individual or small groups of circumferential endings are the fundamental mechanosensitive units of A $\beta$  Field-LTMRs.

How unique is the anatomy of the A $\beta$  Field-LTMR? We next compared the morphological receptive fields of A $\beta$  Field-LTMRs to all other hairy skin LTMRs, including A $\beta$  SA1-LTMRs, A $\beta$  RA-LTMRs, A $\delta$ -LTMRs and C-LTMRs, using sparse genetic labeling strategies specific for each LTMR subtype. Indeed, individual A $\beta$  RA-LTMRs (Figure 7B) associate with many fewer hair follicles (Figure 7F, S6C and S6E) and most of these neurons innervate a small skin area (Figure 7F, S6D and S6F) compared to A $\beta$  Field-LTMRs (Figure 7A and 7F). Individual A $\beta$  SA1-LTMRs typically innervate only one or two touch domes (74% or 26%, respectively) (Figure 7C and 7F), consistent with other measurements (Lesniak et al., 2014; Wu et al., 2012). The remaining hairy skin LTMRs, including A $\delta$ -LTMRs and C-LTMRs, have both smaller morphological receptive fields and innervate fewer hair follicles (Figure 7D-7F, and S6C-S6F). Thus, A $\beta$  Field-LTMRs have unusually expansive receptive fields containing the largest number of specialized mechanosensory terminals of any known mammalian LTMR type.

Given the A $\beta$  Field-LTMR's sensitivity to gentle skin stroke, we next asked whether the expansive terminal fields of A $\beta$  Field-LTMRs exhibit morphological features consistent with

subthreshold integration across multiple circumferential endings. We therefore determined the length of unmyelinated axon segments spanning between the A $\beta$  Field-LTMR spike initiation site (SIS) and its circumferential terminals, and the findings were compared to those of A $\beta$  SA1-LTMRs and A $\beta$  RA-LTMRs. SISs were visualized using myelin basic protein (MBP) in combination of  $\beta$ IV-Spectrin immunostaining (Lesniak et al., 2014; Yang et al., 2007). The average distance between sensory terminals and the SIS was calculated for individual A $\beta$ -LTMRs that were either genetically labeled (A $\beta$  RA-LTMRs or A $\beta$  Field-LTMRs) or immuno-labeled (A $\beta$  SA1-LTMRs). Our analysis showed that SISs and initial myelination segments are uniformly localized within very close proximity to the sensory terminals of both A $\beta$  SA1-LTMRs and A $\beta$  RA-LTMRs (Figure 7H-7J). In contrast, A $\beta$  Field-LTMR SISs are located at variable distances (Figure 7G and 7J) from their circumferential terminals and their unmyelinated axon segments are significantly longer than those of A $\beta$  SA1-LTMRs and A $\beta$  RA-LTMRs (Figure 7J and S7C-S7E). In some cases, A $\beta$  Field-LTMR SISs were located too far from the hair follicle to identify (Figure S7D, data not shown). Similar findings were obtained in experiments in which the three types of A $\beta$ -LTMRs were immuno-labeled using NFH (Figure S7A-S7B). Thus, A $\beta$  Field-LTMRs are unique among hairy skin A $\beta$ -LTMR subtypes in that their initial sites of myelination and SISs are often localized far from the circumferential endings around hair follicles. Taken together, A $\beta$  Field-LTMRs have large, spotty receptive fields containing many weakly mechanosensitive circumferential endings distributed across a large area of skin and SISs located at variable and often considerable distances from the hair follicles with which they associate. We propose that these unique physiological and morphological properties underlie the sensitivity of A $\beta$  Field-LTMRs to gentle stroking across large fields of skin.

## Discussion

Sensory neurons that innervate the skin exhibit tremendous anatomical diversity. For the LTMRs, which mediate our sense of touch, the specialized anatomy of their endings is intimately related to their functions as mechanotransducers. Here we report the generation of murine genetic tools that label A $\beta$  Field-LTMRs, a major population comprising more than 4% of DRG neurons in the mouse. A $\beta$  Field-LTMRs are prevalent across mammalian species, and we find that in mouse each innervates up to 180 hair follicles with specialized NFH<sup>+</sup> circumferential endings. Electrophysiological recordings demonstrate that they are highly sensitive to gentle stroking of the skin but do not respond to hair deflection. Moreover, A $\beta$  Field-LTMRs are increasingly responsive to skin indentation in the noxious range and have large, spotty indentation receptive fields; these properties are unique amongst the A $\beta$ -LTMR subtypes and have been ascribed to A $\beta$  nociceptors (Djoughri and Lawson, 2004). Therefore, A $\beta$  Field-LTMRs may have a nociceptor function, and a test of this possibility will require activation, silencing or ablation approaches and behavioral measures.

The A $\beta$  Field-LTMR is highly sensitive to gentle skin stroking while, on the other hand it is remarkably insensitive to innocuous skin indentation and completely unresponsive to hair deflection. Our *in vivo* recordings, FEM, and morphological analyses support a model in which A $\beta$  Field-LTMRs attain this unique sensitivity through integration across a large number of weakly mechanosensitive hotspots distributed across skin. Although the precise number of hotspots is difficult to ascertain, their approximate number and distribution is

strikingly similar to those of circumferential endings from individual A $\beta$  Field-LTMRs, visualized by sparse genetic labeling. The simplest interpretation of this observation is that each A $\beta$  Field-LTMR circumferential ending is a weakly mechanosensitive unit.

Why are individual circumferential endings of A $\beta$  Field-LTMRs weakly mechanosensitive compared to A $\beta$  RA-LTMR lanceolate endings? One important clue stems from the observation that A $\beta$  RA-LTMR lanceolate endings are closely associated with hair follicle epithelial cells, whereas A $\beta$  Field-LTMR circumferential endings are not. In fact, the naked axonal membrane of individual lanceolate projections resides within 100 nm of the hair follicle epithelial cell membrane (Li and Ginty, 2014) (Figure 5A and 5C). This intimate physical apposition strongly implicates the lanceolate membrane immediately adjacent to the hair follicle epithelial cell as the site where hair deflection-induced mechanical forces are transduced into A $\beta$  RA-LTMR axonal membrane depolarization and excitation. In contrast, A $\beta$  Field-LTMR circumferential endings are located about 4  $\mu$ m from hair follicle epithelial cells where they are enveloped by terminal Schwann cell processes. Another distinguishing feature is that A $\beta$  RA-LTMR lanceolate endings are arranged within an inner follicle region containing longitudinally oriented collagen fibrils, whereas A $\beta$  Field-LTMR circumferential endings are located within an outer follicle region comprised of circumferentially oriented collagen fibrils. Our FEM simulations suggest that these features render A $\beta$  Field-LTMRs less susceptible to strain following both hair deflection and skin indentation.

Our analysis of MBP and  $\beta$ IV-Spectrin localization patterns suggests an additional defining feature of A $\beta$  Field-LTMRs. We found that A $\beta$  Field-LTMR SISs are often distant from their circumferential endings, whereas, in all cases analyzed, A $\beta$  RA-LTMR and A $\beta$  SA1-LTMR SISs are located immediately adjacent to their sensory terminals. Together, these findings support a model in which three distinguishing features of A $\beta$  Field-LTMRs render these neurons selective for stroke among innocuous touch stimuli. These features are: 1) A $\beta$  Field-LTMR circumferential endings are located in an outer layer of circumferentially oriented collagen fibers and thus, compared to A $\beta$  RA-LTMR lanceolate endings, circumferential endings are relatively insensitive to hair deflection or innocuous skin indentation; 2) The terminal fields of A $\beta$  Field-LTMRs are extremely large with individual neurons innervating up to 180 hair follicles in areas that encompass  $\sim 3$  mm<sup>2</sup>, enabling convergence and summation of receptor potentials emanating from multiple circumferential endings; and 3) A $\beta$  Field-LTMR SISs are typically located more than 100  $\mu$ m away from the circumferential endings, suggesting a lower transformation efficiency of receptor potentials to action potentials as well as integration across multiple terminals in the receptive field. We propose that when a “field” of weakly mechanosensitive circumferential endings is activated, receptor potentials emanating from individual circumferential endings converge or summate at distally located SISs to initiate A $\beta$  Field-LTMR spiking. In contrast, the extremely close apposition of hair follicle epithelial cells and A $\beta$  RA-LTMR lanceolate endings and thus the high sensitivity to shear on these endings during follicle movement, as well as the immediate adjacency of these lanceolate endings to SISs enable A $\beta$  RA-LTMRs to spike following deflection of a single hair. Thus, the unique physiological response properties of A $\beta$  Field-LTMRs are a consequence of their terminal morphology, mechanical linkage to the follicle and surrounding skin, and the location of their SISs. It is also likely that LTMRs differ in

expression of mechanically-gated ion channels and transduction machinery, and such molecular differences may also contribute to their unique physiological response properties.

We propose that A $\beta$  Field-LTMRs, with their characteristic expansive receptive fields comprised of large numbers of weakly mechanosensitive circumferential endings, contribute to LTMR activity ensembles that underlie percepts associated with gentle stroking across large fields of hairy skin. Future work will define the contributions of A $\beta$  Field-LTMRs to tactile perception and sensory-motor reflexes, and the postsynaptic partners and circuits in the dorsal horn and brainstem that receive, integrate, and process A $\beta$  Field-LTMR and other LTMR subtype activities.

## Experimental Procedures

Procedures are summarized below; see Supplemental Experimental Procedures for more detailed descriptions of each section.

### Mouse Lines

*TrkC<sup>tdTomato</sup>* and *TrkC<sup>CreER</sup>* mice were generated by introducing a myristoylation signal tagged tdTomato- or CreER-Frt-Neomycine-Frt-loxP cassette into the first coding ATG in exon 1 of the *TrkC* gene (Figure S2A-S2B). Other mouse lines, including the *Brn3a<sup>f(AP)</sup>*, *CGRP $\alpha$ -GFP* and *Npy2r-GFP* BAC transgenic, *Ret<sup>CreER</sup>*, *Ret<sup>CFP</sup>*, *Ret<sup>f(CFP)</sup>*, *R26<sup>LSL-YFP</sup>*, *R26<sup>LSL-tdTomato</sup>* (*Ai9*), *R26<sup>FSF-LSL-tdTomato</sup>* (*Ai65*), *TH<sup>CreER</sup>*, *TrkB<sup>CreER</sup>* mouse lines, have been described previously (Badea et al., 2009; Gong et al., 2003; Li et al., 2011; Luo et al., 2009; Madisen et al., 2015; Rotolo et al., 2008; Rutlin et al., 2014; Uesaka et al., 2008).

### Histological Analyses

Immunohistochemistry of tissue sections, whole mount immunohistochemistry, and whole mount AP histochemistry were performed using standard procedures (Li et al., 2011).

### Electrophysiological Recordings

*In vivo* recordings were performed from genetically identified L4 DRG neurons in isoflurane anesthetized mice. To measure stimulus-evoked action potentials, a borosilicate extracellular electrode was visually guided under reflective optics to fluorescently labeled neurons and a loose seal formed. Stimuli were delivered to the hairy skin of the dorsal hindlimb. Intracellular recordings using an *ex vivo* skin-nerve preparation of 2-3 months old animals were performed with dorsal back skin, dorsal cutaneous nerve, thoracic DRGs and spinal cord intact (Woodbury et al., 2001).

### Electron Microscopy

Fixed hairy skin was stained with 1% osmium tetroxide/1.5% potassium ferrocyanide followed by 1% uranyl acetate, gradually dehydrated with ethanol, and embedded in Epon for ultrathin sectioning and TEM imaging.

## FEM simulations

FEM simulations of the hair follicle ultrastructure were built in Abaqus Standard (ver. 6.13). The model includes five parts: the skin, the hair core, the surrounding hair follicle, and two layers containing collagen fibers with co-axial and circumferential orientations.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

## Acknowledgements

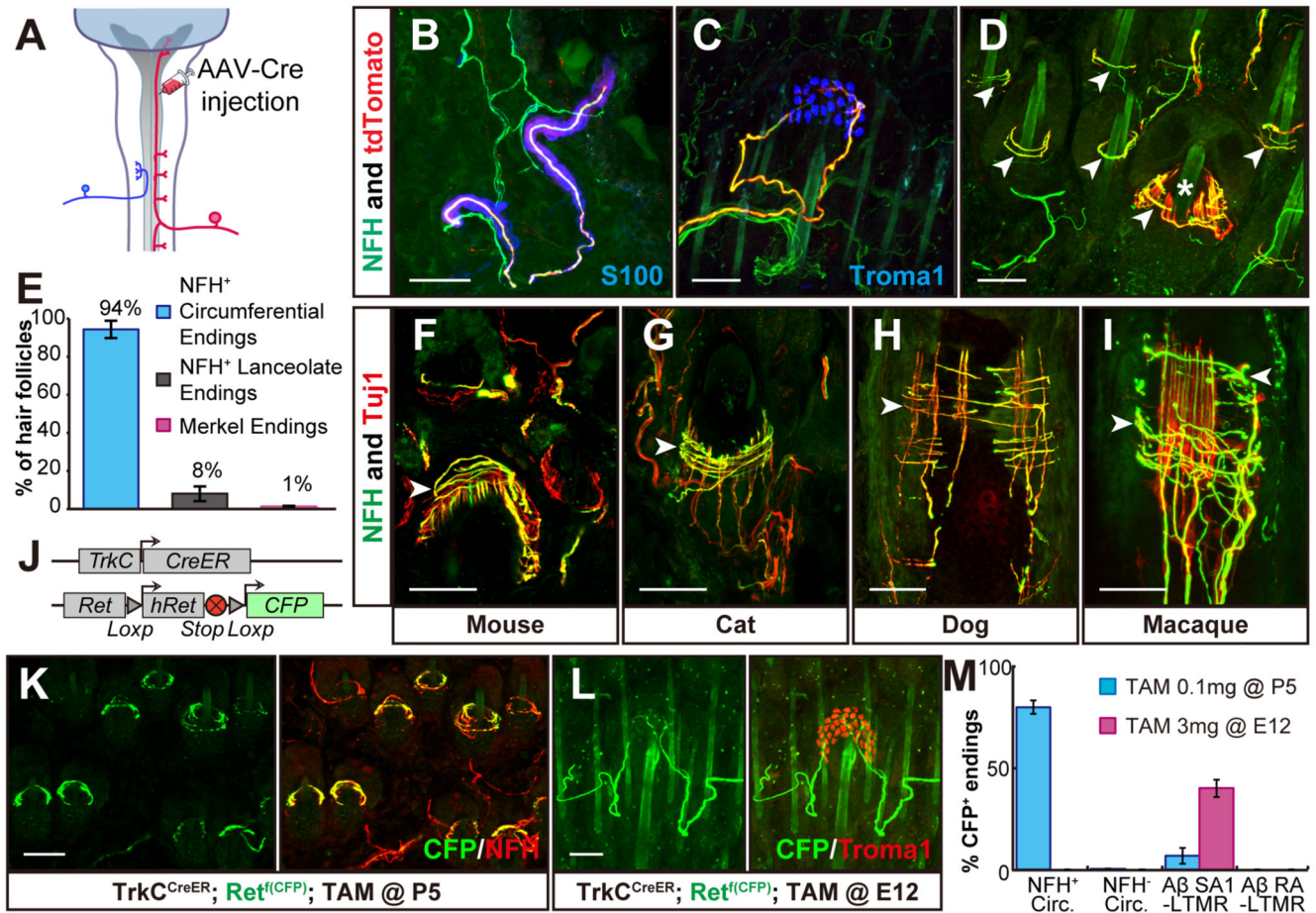
We thank Michael Rutlin for help with targeting vector design, Lishi Li and Steven Hsiao for the macaque skin images, Hideki Enomoto for *Ret<sup>fl(CFP)</sup>* and *Ret<sup>CFP</sup>* mice, Jeremy Nathans for *Brn3a<sup>fl(AP)</sup>* mice, Matthew Rasband for the  $\beta$ IV-Spectrin antibody, Tom Ferstl for tissue samples, and Emily Kuehn for AAV virus characterization. We are grateful to Robert LaMotte for assistance with the *in vivo* physiological recording preparation, Rachel Wilson for use of the Laser Doppler Vibrometer, Maria Erissson (Harvard EM core facility), Ciara Bolger and Wei-Chung Lee for assistance with the EM analysis, and Matthew Pecot, Zhang Yilei and members of the Ginty laboratory for valuable comments on the manuscript. This work was supported by NIH grants T32 NS007484-14 (BPL), F32 NS095631-01 (BPL), NS44049 (CJW), DE022750 and NS34814 (DDG). DDG is an investigator of the Howard Hughes Medical Institute.

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**Figure 1. Genetic labeling of sensory neurons that innervate the dorsal column nuclei**

(A-D) The C1 DC of *R26<sup>LSL-tdTomato</sup>* mice was injected with AAV2/1-Cre virus to retrogradely label neurons that project to the DCN. (B-D) In hairy skin, at least 4 types of terminals are labeled by tdTomato: an unknown ending type innervating S100<sup>+</sup> terminal organs that resemble Pacinian corpuscles (B), Merkel endings from A $\beta$  SA1-LTMRs that associate with Troma1<sup>+</sup> Merkel cells (C), lanceolate endings that are NFH<sup>+</sup> ( $97.8 \pm 1.2\%$ , 156 terminals,  $n = 3$ ) and derived from A $\beta$  RA-LTMRs (D, asterisk), as well as circumferential endings that are also NFH<sup>+</sup> ( $99.5\% \pm 0.5\%$ , 465 terminals,  $n = 3$ ) (D, arrowhead). (E) NFH<sup>+</sup> circumferential endings, NFH<sup>+</sup> lanceolate endings, and Merkel endings innervate  $94 \pm 3\%$ ,  $8 \pm 2\%$ , and  $0.8 \pm 0.2\%$  of hair follicles, respectively (3964 hair follicles,  $n = 3$ ). (F-I) Immunostaining of NFH and Tuj1 on hairy skin sections reveals that NFH<sup>+</sup> circumferential endings (arrowhead) can be found in mouse (F), cat (G), dog (H), and macaque (I). (J) Diagram of the TrkC and Ret intersectional genetic labeling strategy. (K) Double immunostaining of hairy skin sections from *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* mice treated with 0.1 tamoxifen at P5 reveals that CFP specifically labels the majority of NFH<sup>+</sup> circumferential endings ( $79.6 \pm 3.3\%$ , 1142 terminals,  $n = 4$ ) and all of CFP<sup>+</sup> circumferential endings are NFH<sup>+</sup> ( $99.2 \pm 0.2\%$ , 904 terminals,  $n = 4$ ). (L) Whole-mount immunostaining of hairy skin from *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* mice treated with 3mg tamoxifen at E12.5 reveals that CFP specifically labels A $\beta$  SA1-LTMRs innervating Troma1<sup>+</sup> Merkel cells ( $40 \pm 4\%$ , 78

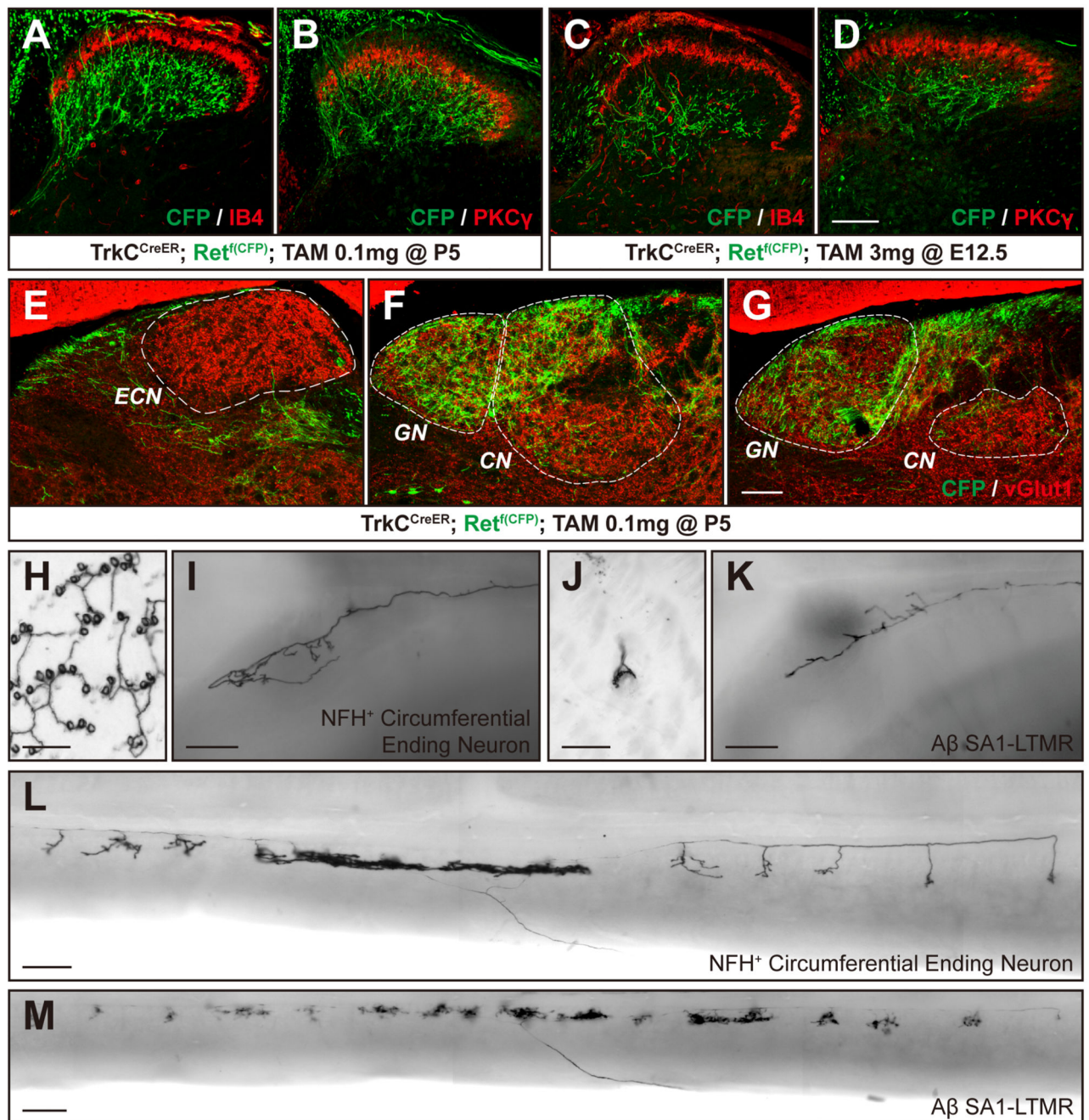
terminals,  $n = 3$ ). (M) Quantification of the percentage of sensory endings labeled by CFP in K-L. Scale bar: 50  $\mu\text{m}$  (B-D, F-I, K-L). See also Figure S1, S2, and S3.

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**Figure 2. Neurons with NFH<sup>+</sup> circumferential endings innervate the deep spinal cord dorsal horn as well as the DCN**

(A-D) Spinal cord sections from *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* mice treated with 0.1 mg tamoxifen at P5 or 3 mg tamoxifen at E12.5. The central projections of neurons with NFH<sup>+</sup> circumferential endings labeled by CFP are located in lamina II<sub>iv</sub> through IV (n = 9), ventral to lamina II<sub>id</sub> labeled by IB4 (A) but partially overlapping with lamina II<sub>iv</sub> labeled by PKC $\gamma$  (B). The central projections of A $\beta$  SA1-LTMRs labeled by CFP are located in lamina III through V (n = 4), ventral to lamina II<sub>id</sub> and lamina II<sub>iv</sub> labeled by IB4 (C) and PKC $\gamma$  (D). (E-G) Transverse brainstem sections from *TrkC<sup>CreER</sup>; Ret<sup>f(CFP)</sup>* mice treated with 0.1 mg

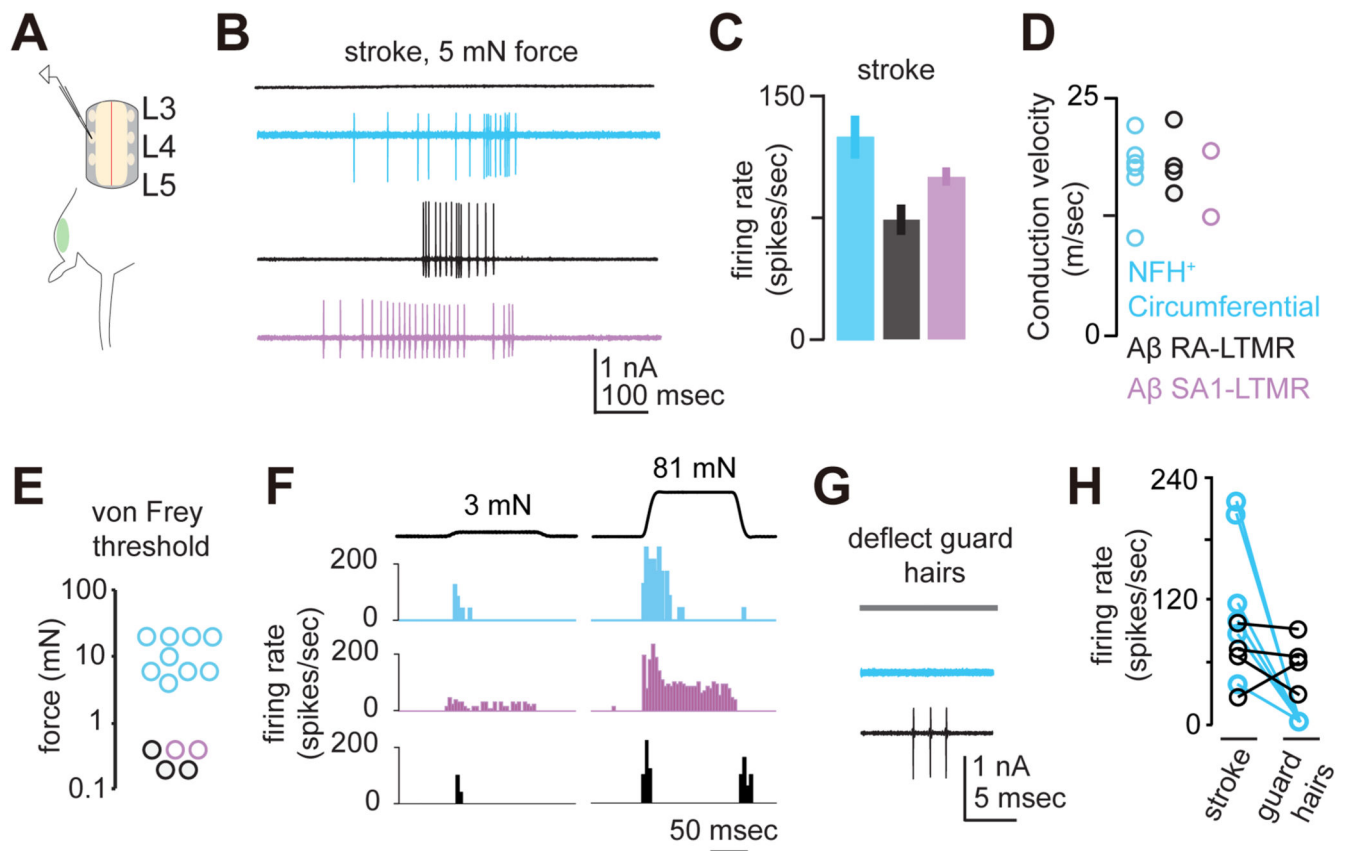
tamoxifen at P5 reveals the innervation pattern in the DCN, which is marked by vGlut1. (H-M) Whole-mount AP staining of the skin (H and J), brainstem (I and K), and spinal cord (L and M) from *TrkC<sup>CreER</sup>; Brn3a<sup>f(AP)</sup>* mice treated with 0.001mg tamoxifen at P8 reveals the peripheral terminals and central projections of neurons with circumferential endings (H, I and L) or Merkel endings (J, K, and M). Dorsal view of the left DCN and left thoracic spinal cord reveals that axons from a single labeled circumferential ending neuron and A $\beta$  SA1-LTMR project into the spinal cord and form collaterals in both the rostral and caudal directions (L and M). The rostral end of this axon extends to the DCN, where it forms collaterals (I and K). Scale bar: 100  $\mu$ m (A-G) and 200  $\mu$ m (H-M). See also Figure S4.

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**Figure 3. Neurons with NFH<sup>+</sup> circumferential endings are A $\beta$  Field-LTMRs**

(A) The targeted *in vivo* DRG recording preparation. (B) Representative responses of A $\beta$  SA1-LTMR, A $\beta$  RA-LTMR, and NFH<sup>+</sup> circumferential ending neurons to skin stroke delivered by a force controlled brush translated by an ultrasonic piezoelectric stage. The recorded holding force (black, ~5 mN) is shown at top. The stroke speed is 10 mm/sec. (C) Quantification of stroke-evoked action potential firing rates in A $\beta$  SA1-LTMR (n = 4), A $\beta$  RA-LTMR (n = 5), and NFH<sup>+</sup> circumferential neurons (n = 10). Responses were computed as the average of the three highest firing rates observed in a 100 msec window which corresponds to the approximate time required for the brush to transit the full arbor of an A $\beta$ -LTMR. NFH<sup>+</sup> circumferential ending neurons responded to innocuous stroke with firing rates indistinguishable from those of other hairy skin LTMRs (p = 0.27 Welch's t-test). (D) The conduction velocities of NFH<sup>+</sup> circumferential ending neurons are indistinguishable from those of other hairy skin A $\beta$  RA- and SA-LTMRs recorded (p = 0.871, Welch's t-test). (E) Von Frey threshold measurements reveal that the force thresholds of NFH<sup>+</sup> circumferential ending neurons are higher than those of other A $\beta$  subtypes (90% bootstrap CI [21.2, 52.3] fold higher). (F) Temporal patterns of responses to indentation. Representative peristimulus time histograms (PSTHs) show that NFH<sup>+</sup> circumferential neurons adapt more rapidly than A $\beta$  SA1-LTMRs but lack the pronounced off-step responses of the A $\beta$  RA-LTMRs. The indentation force recording (top) is aligned to the PSTH showing the mean firing rate computed across 15 stimulus presentations (7 msec bin widths). (G) Representative spiking responses of NFH<sup>+</sup> circumferential ending neurons and A $\beta$  RA-

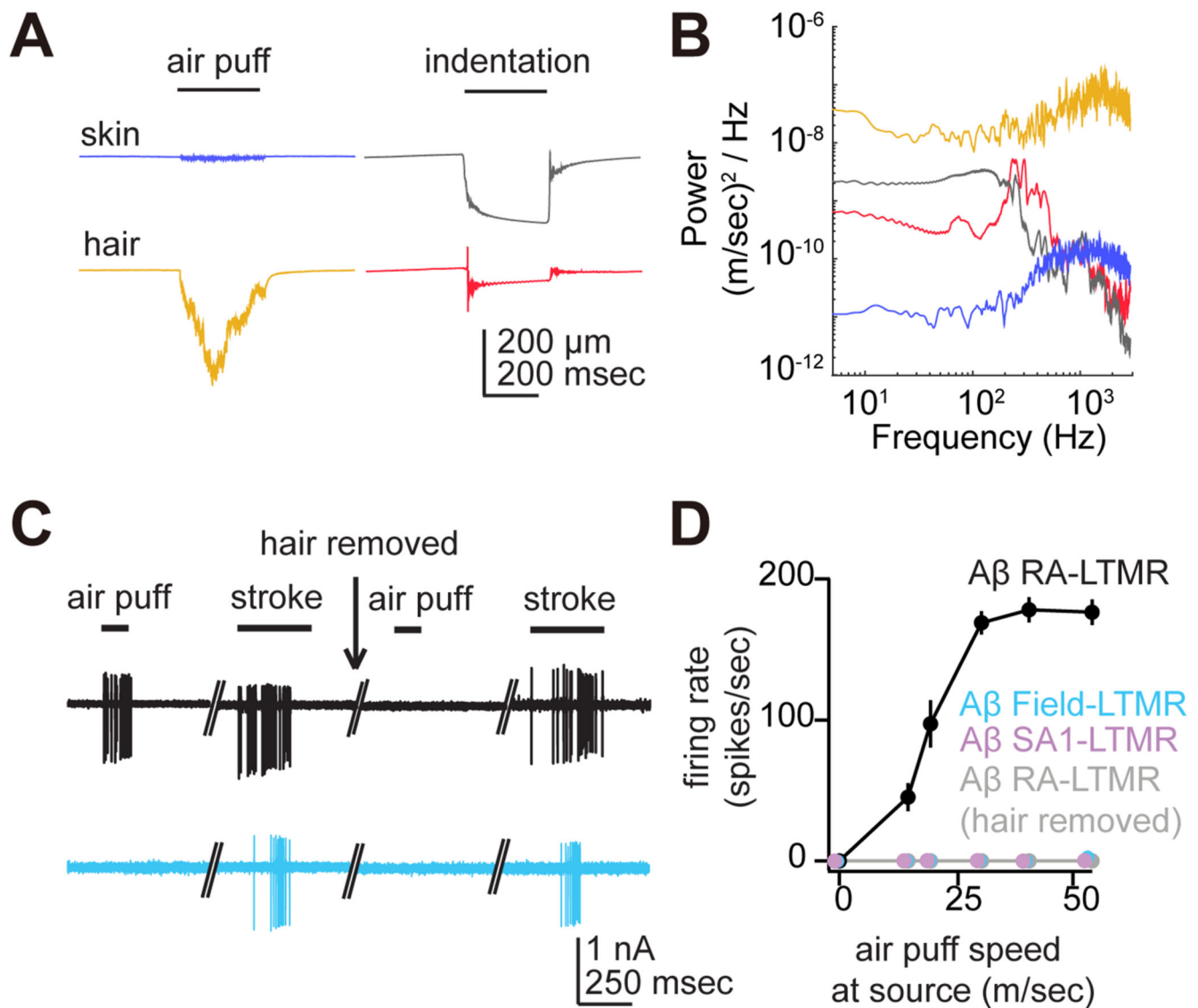
LTMR to deflection of guard hairs. (H) Group data showing that all A $\beta$  RA-LTMRs respond similarly to deflection of guard hairs and skin stroke ( $p = 0.85$ , paired t-test), while NFH<sup>+</sup> circumferential ending neurons are insensitive to guard hair deflection ( $p = 0.017$ , paired t-test). As in (C), evoked firing rates are computed over a 100 msec window. See also Figure S5.

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**Figure 4. A $\beta$  Field-LTMRs are insensitive to hair deflection yet sensitive to direct stimulation of skin**

(A) Laser Doppler Vibrometric measurements of skin and hair movement in response to air puff (left, 15m/sec at the source) and indentation (right, 2g von Frey filament). Shown are representative displacements of the base of a hair shaft and a nearby patch of skin. (B) Group data showing the power spectral density averaged across stimulus presentations and measurement locations ( $n = 5$  hairs,  $n = 5$  skin locations). For air puff, the motion recorded at the base of hairs exceeds that of the surrounding skin by three orders of magnitude. (C) Representative *in vivo* recordings of A $\beta$  Field-LTMR (blue) and A $\beta$  RA-LTMR (black) responses to air puff and stroke, before and after hair removal. The recording was maintained continuously in all cases. (D) Air puff evoked spiking responses from A $\beta$  Field-LTMRs ( $n = 6$ ), A $\beta$  SA1-LTMR ( $n = 3$ ), A $\beta$  RA-LTMRs ( $n = 4$ ) and A $\beta$  RA-LTMR after hair removal ( $n = 4$ ). Firing rates computed over the entire 100 msec air puff are plotted against the speed of the air at the stimulator nozzle, which was placed 12mm above the skin. All A $\beta$

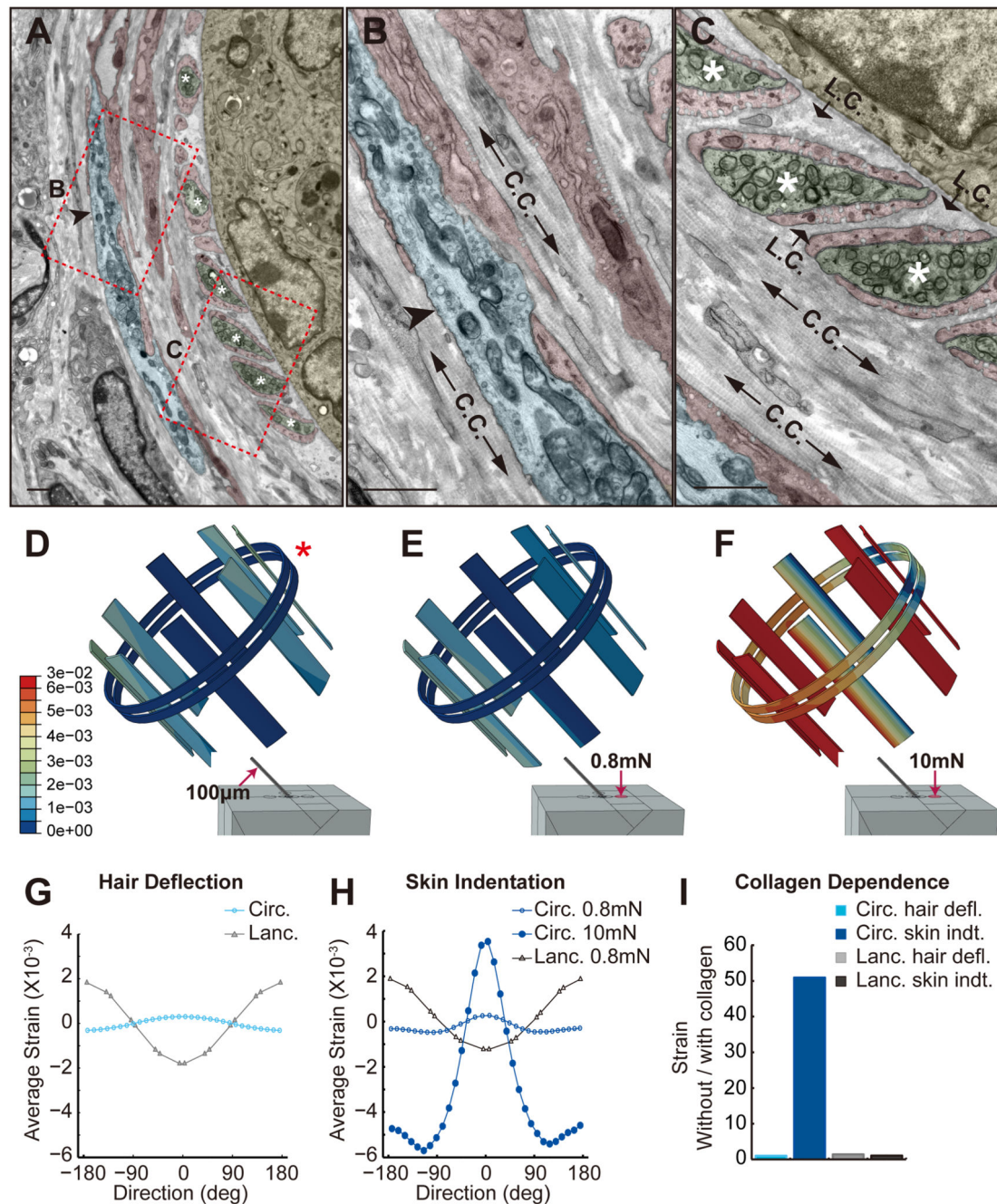
RA-LTMR retained sensitivity to stroke after hair removal (C, data not shown), demonstrating that they remained mechanically sensitive despite loss of air puff responses. Traces are offset for clarity.

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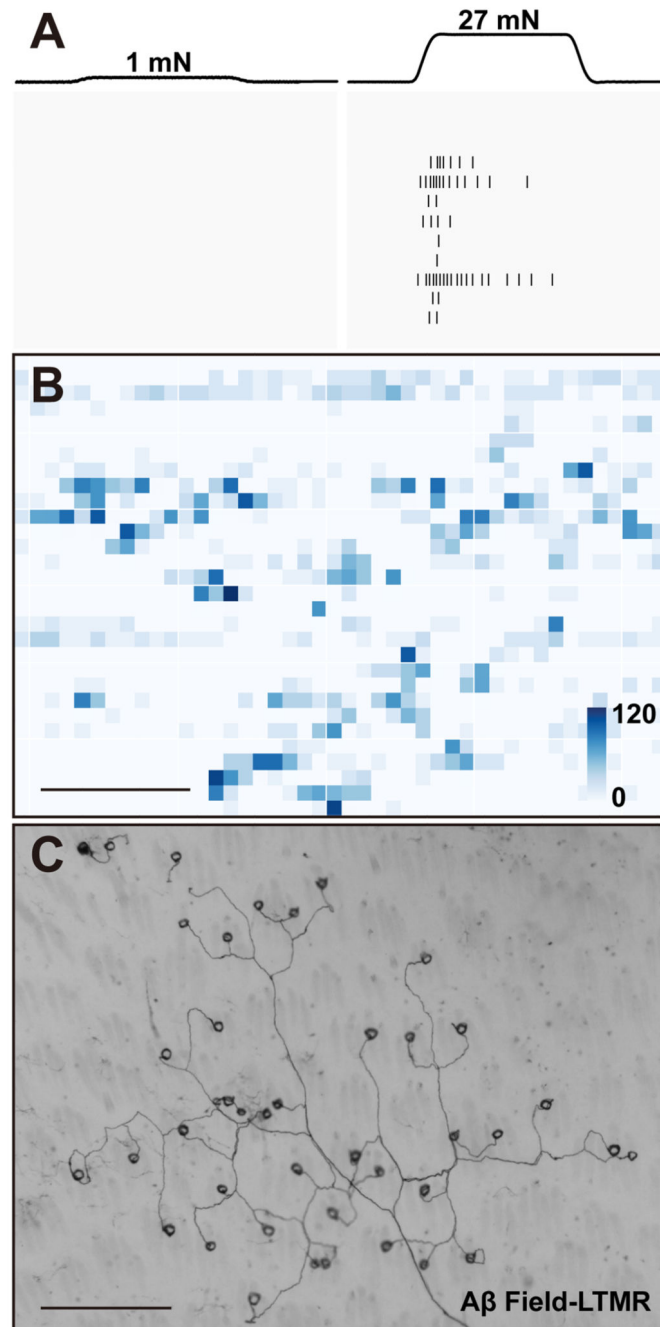
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**Figure 5. Ultrastructural features of lanceolate and circumferential endings that support the distinct response properties of  $A\beta$  RA-LTMRs and  $A\beta$  Field-LTMRs**

(A-C) TEM images of cross-sections through a lanceolate and circumferential ending complex at a hair follicle. (B-C) Magnified view of the region in the red box of (A). The lanceolate ending (white asterisks) and terminal Schwann cell complexes are embedded in longitudinal oriented collagen fibers (L.C.) in close proximity to hair follicle epithelial cells. Circumferential endings (arrowheads) are embedded in circumferentially oriented collagen fibers (C.C.) in the outer layer. The lanceolate endings, circumferential endings, terminal Schwann cells, and hair follicle epithelia cells are pseudo colored in green, blue, and pink,

and yellow, respectively. (D-I) The FEM simulations of the strain acting on lanceolate or circumferential endings in response to mechanical stimuli. (D-F) Top: Contour plots of strain distribution along the lanceolate and circumferential endings in response to 100  $\mu\text{m}$  hair deflection (D), 0.8 mN skin indentation (E), and 10 mN skin indentation (F). Bottom: schematic diagram of the stimulations. (G-H) Plot of the average strain along different positions of lanceolate or circumferential terminals. The zero location is marked by the red asterisk in Figure 6D. (I) The ratio of accumulated strain in the absence and the presence of collagen layers along lanceolate or circumferential endings under either 100  $\mu\text{m}$  hair deflections or 0.8mN skin indentations. Scale bar: 1  $\mu\text{m}$  (A-C). See also Supplemental Table 1.



**Figure 6. A $\beta$  Field-LTMRs have large receptive fields comprising many weak mechanosensitive endings**

(A) *In vivo* loose patch electrophysiological recordings from an A $\beta$  Field-LTMR in response to 1 mN (left) and 27 mN (right) spatially patterned skin indentations with a 20  $\mu$ m tipped tungsten probe. The indenter was positioned over the skin in 50  $\mu$ m steps, such that adjacent steps correspond to adjacent rows in the raster plot. (B) Receptive field of an A $\beta$  Field-LTMR mapped with a 27 mN indentation. Grid indentations of A $\beta$  Field-LTMRs reveal punctuated “hotspots” that extend for 50-100  $\mu$ m and are separated by insensitive patches (n

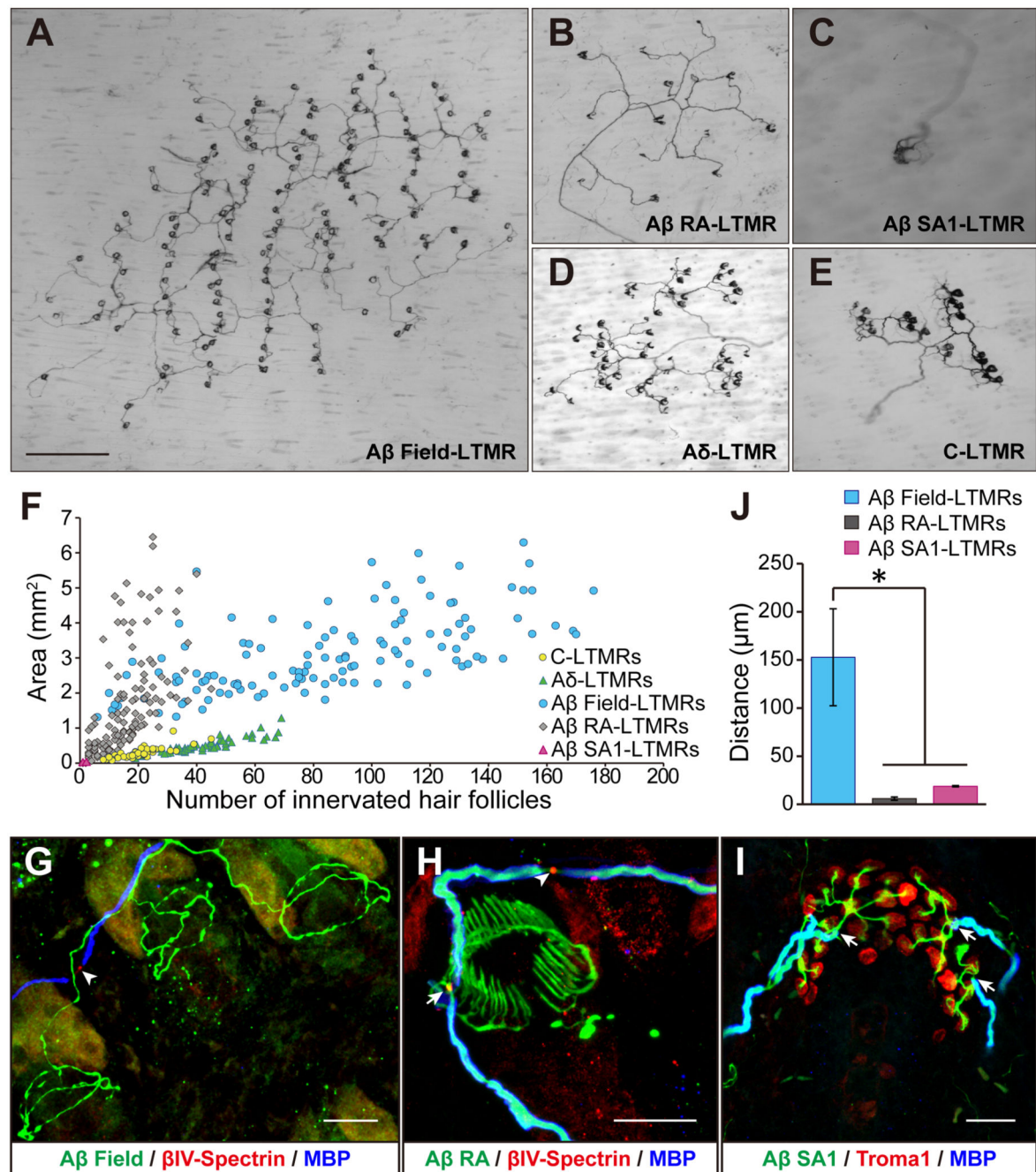
= 2). (C) Whole-mount AP staining of hairy skin reveals peripheral terminals from individual A $\beta$  Field-LTMRs. Scale bar: 500  $\mu$ m (B-C). See also Figure S6.

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**Figure 7.  $A\beta$  Field-LTMRs have large terminal fields and a long distance between their axon terminals and SISs**

(A-E) Whole-mount AP staining of hairy skin reveals peripheral terminals from individual LTMR subtypes sparsely labeled by a *Brn3a*<sup>(AP)</sup> reporter line. (F) Quantification of the number of innervated hair follicles and terminal field area from different LTMR subtypes including  $A\beta$  Field-LTMRs (n = 125, 8 mice),  $A\beta$  RA-LTMRs (n = 127, 10 mice),  $A\beta$  SA1-LTMRs (n = 31, 16 mice),  $A\delta$ -LTMRs (n = 55, 4 mice), as well as C-LTMRs (n = 41, 3 mice). Each dot represents a single neuron. (G-I) Whole-mount immunostaining of hairy skin reveals the relationship between MBP, the SIS as inferred from  $\beta$ IV-Spectrin and the

terminals of three A $\beta$ -LTMRs subtypes. Arrows or arrowheads point out the SISs or nodes of Ranvier, respectively. Note that the red  $\beta$ IV-Spectrin puncta in G (arrowhead) is associated with a different myelinated axon that is not belong to the genetically labeled A $\beta$  Field-LTMRs. (J) Quantification of the distance of non-myelinated axons reveals that A $\beta$  Field-LTMRs have longer non-myelinated axons ( $153 \pm 50$ ,  $n = 5$ ) compared to A $\beta$  RA-LTMRs ( $6.0 \pm 1.8$ ,  $n = 12$ ) ( $p = 0.03$ , student t-test) and A $\beta$  SA1-LTMRs ( $18.8 \pm 0.7$ ,  $n = 8$ ) ( $p = 0.04$ , student t-test). Scale bar: 500  $\mu$ m (A-E). 20  $\mu$ m (G-I). See also Figure S6 and S7.